Kaya: one twin per patient

Open + synthetic data only · Decision support

It shows what glucose does to the liver, eyes and kidneys, and lets you test a change before trying it on the real patient.

1 WHAT GOES IN

- Wearable stream
- EHR and labs
- Retina scan
- Patient quick log
- Voice or chat

2 THE KAYA TWIN

Patient state encoder
one fused twin

Glucose
2-hour forecast and alerts

Liver
fat and fibrosis risk

Eye
retinopathy grade and path

Kidney
eGFR path and risk

What-if engine
change a meal, walk, sleep or medicine
see the new glucose curve and organ risk

Explainable · calibrated uncertainty · safety rules in code · served by one REST API

3 WHO USES IT, AND HOW

PATIENT
on a phone, in 4 languages

1. **Today** — one next action, and why
2. **Quick log** — medicines, walk, mood
3. **Try a change** — see tonight's curve
4. **Ask** — by voice or text

the log becomes the doctor's adherence diary

same twin, same what-if engine

DOCTOR
triage, understand, decide

1. **Triage** — sorted by risk, with reasons
2. **Full record** — glucose, organs, labs
3. **What-if studio** — test before the real patient
4. **PDF summary** — reports with trends

HOSPITAL IT — Plugs in through the REST API: live readings in · FHIR R4 out · Docker deploy. Governance and ABDM are on the roadmap.

Kaya is decision support for clinicians, not a diagnostic device. All patients are synthetic. Models, metrics and frameworks are on the next page.

Under the hood

The models, what they were trained and tested on, and the software around them.

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MODELS AND EVIDENCE

Glucose

GRU (PyTorch) + LightGBM ensemble

Quantile + adverse-event heads
Conformal 80% bands, isotonic-calibrated alerts
Trained on real CGMacros + synthetic cohort
RMSE 12.6 / 21.1 / 28.2 mg/dL at 30 / 60 / 120 min
Catches 72% of lows, median 45 min ahead

Liver

LightGBM + logistic regression, SHAP

NHANES 2017-18, 4,745 adults, FibroScan labels
Advanced fibrosis AUROC 0.83 (FIB-4: 0.63)
Steatosis AUROC 0.83 (FLI: 0.82)
Liver-fat state: mechanistic, driven by glucose

Eye

EfficientNet-B2 (timm) + Grad-CAM

Ordinal regression on APTOS 2019
Kappa 0.89 on APTOS; 0.71 on Indian IDRiD
after fine-tuning (0.57 without)
2-year progression risk: simulation

Kidney

LightGBM + logistic regression, SHAP

NHANES 2017-18, 4,827 adults
Low eGFR AUROC 0.85, albuminuria 0.76
Modest gain over a simple baseline, stated openly
5-year eGFR path: simulation

LANGUAGE AND VOICE AI

Sarvam AI, called from the server only (the key never reaches the browser)

saaras:v4 speech-to-text · sarvam-105b-conversations chat · bulbul:v4-flash text-to-speech
English, Hindi, Telugu, Kannada. Emergency and dose rules run in code, before and after the model

FRAMEWORKS AND DELIVERY

FastAPI, Pydantic, WebSocket, FHIR R4, ReportLab PDFs · React 18 + Vite

Live mode runs the GRU in pure NumPy, about 1 ms per reading, no GPU
pandas, scikit-learn, SHAP, simglucose · Docker Compose, GitHub Actions CI · trained on 3x RTX A5000

HOSPITAL IT: TODAY AND ROADMAP

Today: REST API, live readings, FHIR Patient and Observation, Docker deploy

Roadmap, not built: consent records, access audit log, model-health page,
ABDM alignment, FHIR DiagnosticReport and Condition

All metrics are out-of-sample. The links from glucose patterns to organ outcomes are simulated on a synthetic cohort and are not clinically validated.